

Assessment of Variable Selection Methods using Simulation

Introduction

Regression analysis is a statistical tool used to assess the explanatory role of two or (often) more predictor variables for an outcome variable. In tasks where there are no *a priori* set of predictors to be included in a model, a variable selection analysis/process is undertaken using several approaches such as backwards, forward, stepwise, best subset, or shrinkage/regularization variable selection. To perform selection, these approaches optimize different criterion including Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), adjusted R-squared, Mallows' Cp, and p-value.

In this project, we compare the statistical operating characteristics of (a) backwards selection approaches based on p-value, AIC, and BIC criteria, and (b) regularization approaches [least absolute shrinkage and selection operator (LASSO), and elastic net] based on mean squared error (MSE) criteria, under different sample size scenarios and strength of correlation among predictors. The operating characteristics (or performance metrics) of interest include true and false positive rates, bias, coverage of a 95% confidence interval, and type I & type II errors.

Methods

Simulation setup and data generation: Variable selection approaches including criteria were compared under 6 key scenarios: 2 sample size scenarios of 250 and 500, and within each sample size, 3 correlation coefficient (ρ) magnitudes of 0.00, 0.35, and 0.70 corresponding to the exchangeable correlation among predictors. A positive non-zero ρ induces multicollinearity among predictors, and with exchangeability, ρ between any 2 pair of variables is the same. For each scenario, 20 predictors were simulated; the first 5 predictors have betas/coefficients 0.5/3, 1/3, 1.5/3, 2.0/3, and 2.5/3, while the other 15 predictors have betas of 0 and are therefore not

associated with the outcome variable. The outcome and predictor variables were simulated from the normal distribution, and data was generated using the `hdrm::gen_data()` function in R.

Data analysis: The variable selection approaches (and optimized criteria) include (a) backwards selection (p-value), (b) backwards selection (AIC), (c) backwards selection (BIC), (d) LASSO [$\alpha=1$] (MSE) with penalty terms $\lambda = 'min'$ and $\lambda = '1se'$, (e) elastic net [$\alpha=0.5$] (MSE) with $\lambda = 'min'$ and $\lambda = '1se'$, (f) elastic net [$\alpha=0.95$] (MSE) with $\lambda = 'min'$ and $\lambda = '1se'$, and (g) Oracle (truth) model. The penalty term controls model complexity. Each of these approaches were fitted in a multivariable linear regression framework using R Statistical software. Approach (a) was fitted using `olsrr::ols_step_backward_p()` with a p-value significance criterion of 0.05. Approaches (b) and (c) were fitted using `stats::step()` with a penalty of 2 and N (sample size) which corresponds to AIC and BIC criteria respectively. Approaches (d), (e), and (f) with $\alpha = 1, 0.5$, and 0.95 respectively were fitted using `glmnet::cv.glmnet()` and MSE was specified as the optimization criteria. For each of these approaches, cross-validated MSE was computed across a grid of λ values using 5-fold cross-validation. For each model, two choices of λ were assessed – the value (of λ) that minimizes the cross-validated MSE ($\lambda = 'min'$) and the largest value (of λ) whose cross-validated MSE is within 1 standard error of the minimum ($\lambda = '1se'$), resulting in a more parsimonious model. Five-fold cross validation was used so that, in the 250-sample size scenario, the ratio of sample size in each fold to number of predictors (20) will be reasonable and result in reliable variable selection. In addition to the elastic net model using $\alpha=0.5$, which applies the same weight to the L1 and L2 terms, a separate elastic net model with $\alpha=0.95$ was fitted. Compared to the first, this model performs more aggressive variable selection, as only 5% weight is placed on the L2 term, making its behavior closely resemble that of a LASSO model. For the shrinkage methods, LASSO and elastic net,

lm () was used for post-selection inference whereby the variables selected by these models were refit and relevant model estimates extracted.

In addition to these 9 modelling approaches (3 backwards selection, and 2 each of LASSO, elastic net [$\alpha=0.5$], and elastic net [$\alpha=0.95$]), a multivariable linear regression model, without variable selection, was fitted using the 5 predictors known to be associated with the outcome. This model serves as the Oracle (truth) model – it includes only the 5 variables known to be associated with the outcome and serves as a reference for the comparison of metrics obtained from the other models. The output of each model was a matrix of 4 columns representing the point estimate, p-value, and lower & upper 95% confidence interval. These estimates are used to calculate performance metrics.

Performance metrics: The performance metrics used to compare the modelling approaches include bias of the coefficient, 95% coverage, rejection, true positive (TP), and false positive (FP) rates. All these metrics were estimated unconditionally, that is, estimated using the entire simulation runs and not restricted to only the runs where the predictors of interest were selected. Bias is defined as the difference between the coefficient estimate and the true value; 95% coverage is an indicator function for whether the true coefficient is included in a 95% confidence interval for the estimated coefficient; rejection is defined as whether a variable is selected and the p-value for the estimated coefficient is ≤ 0.05 – under the null model this is type I error and under the alternative, it is power ($1-\beta$ [type II error]); TP and FP is an indicator function for selection of a non-zero and zero coefficient variable respectively. Bias, coverage, rejection (power), and TP estimates were presented for each of the 5 signal variables and 2 noise variables – the noise variables will serve as a representative of the other 15 noise variables, as well as provide an

estimate of type I error. FP (TP) was summarized as the proportion of the 15 noise (5 signal) variables included in a model run.

Simulation runs and reproducibility: The data generation and 5-fold cross-validation steps are based on random sampling, therefore a unique seed value was specified at each run for reproducibility. For each simulation run, data was generated, then analysis of the 10 variable selection methods was performed and each time creating a matrix of model estimates which was used to calculate performance metrics that were stored and then summarized. These simulation steps were repeated for 10,000 Monte Carlo samples using parallel computation to improve efficiency, and during each run, the seed value, simulation scenario settings, and performance metrics were saved for later summarization. Simulation was based on 10,000 MC samples to attain a desired precision of at most 0.5% using the sampling distribution of the binary performance metrics as reference.

Results

The run time for 10,000 Monte Carlo simulations with parallel computation using 10 cores was 3.21 hours.

Figure 1 shows a plot of mean bias for each variable selection by sample size and correlation for 5 true signal (X1, X2, X3, X4, and X5) and 2 noise variables (X6, X11). The oracle model was unbiased independent of correlation coefficient. All methods shows unbiasedness under no correlation ($\rho=0.00$) except for X1 – this is attributed to the small effect size for this variable. Under moderate correlation ($\rho = 0.35$), the backwards selection methods were near unbiased (except for X1). All the shrinkage methods underestimated the true effect (i.e., bias was < 0). Under strong correlation ($\rho=0.70$), the backwards selection method using AIC had the least bias; backwards based on p-value was the next least unbiased. Generally, increasing correlation

increases bias, higher sample size ($n=500$) reduces bias compared to a sample size of 250, and the shrinkage methods performed worse than the backwards selection methods. **Table 1** shows numerical values plotted in Figure 1.

Figure 2 shows the 95% coverage. As expected, coverage is poor for the noise variables.

Coverage is generally at or near the expected value of 95% across the different rho values and sample sizes. However, this is true only for the true signal variables with moderate to large effect sizes, not X1. As described above, higher sample size has better coverage. Under X1, there is no strong discernable impact of correlation on coverage; backwards BIC has the worst coverage compared to other backwards methods. **Table 2** shows the numerical values plotted in Figure 2.

Table 3 shows rejection under the null (using X6 and X11) and under the alternative at different effect sizes (using X1 – X5). All the methods maintained a type I error rate at or below 5% for both sample size and correlation scenarios. Statistical power is at least 97% for the variables with strong effect (X3, X4, and X5) regardless of correlation or sample size. At higher rho ($\rho=0.70$), statistical power is reduced especially for sample sizes of 250, and the shrinkage methods had lower power. Under small effect sizes (X1), there is substantial power loss, and it reduces inversely with correlation. Also, under X1, the positive impact of sample size on power is very apparent.

Table 4 shows TP and FP rates by selection method, sample size, and correlation for X1, X2, X3, X4, X5, the proportion of counts of signal variables selected (with a denominator of 5), and proportion of noise variable selected (with a denominator of 15). With large effect size, each of the signal variables X2, X3, X4, X5 were selected almost 100% of the time, independent of correlation and sample size. Under low effect size (X1), TP is higher with a higher sample size of 500 vs 250. At higher correlation, the shrinkage methods have higher TP than the backwards

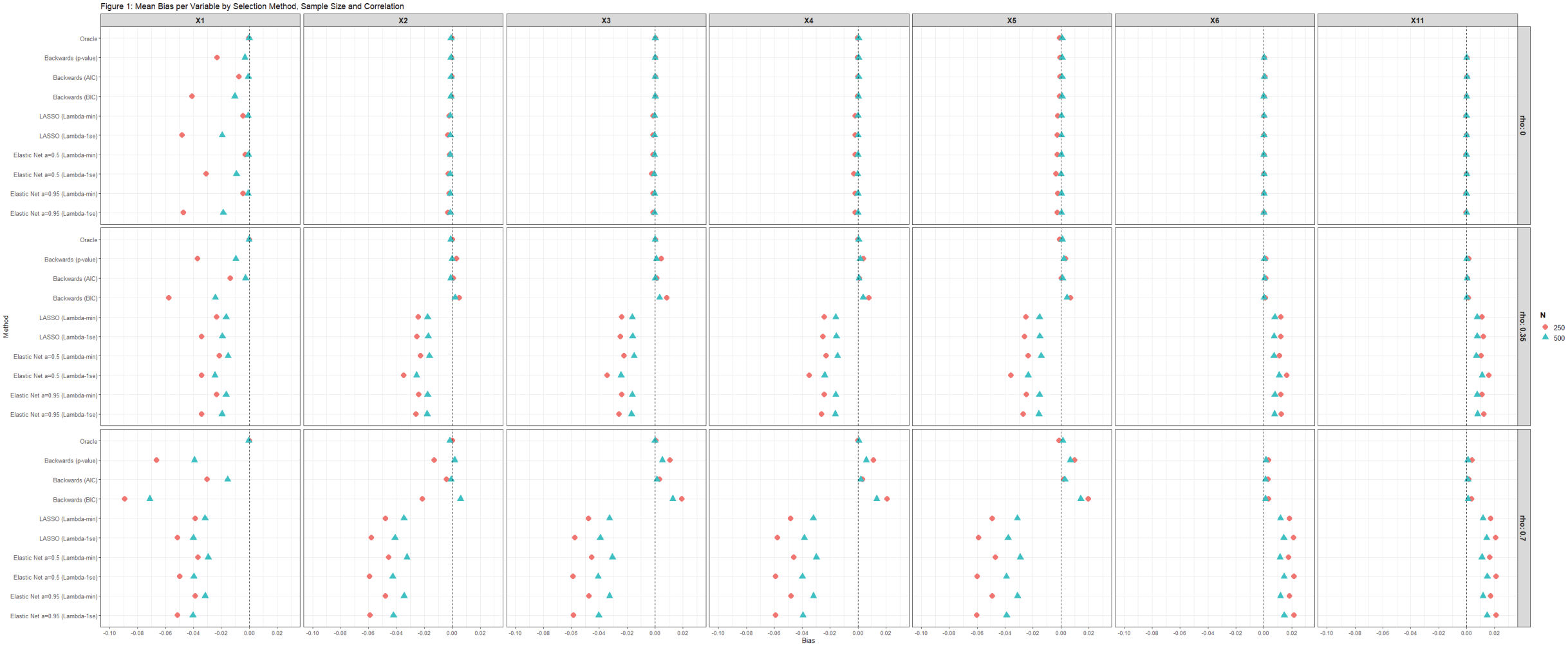
selection methods. Overall, the true signal variables are selected a higher proportion of the time, at least 92% when correlation is 0.00 and 0.35 with sample size of 250, and at least 95% of the time when sample size is 500, (see **Prop. X1 – X5** column in Table 4). There is no discernable pattern for FP rates with increasing correlation or sample size (see **Prop. X6 – X20** column in Table 4) but the backwards selection approaches have a lower FP overall, and within the shrinkage methods, using $\lambda = 1se$ reduces FP rates which is appropriate since this is more parsimonious model compared to using $\lambda = \min$.

The elastic net with $\alpha=0.95$ performs very similarly to the LASSO across board for the performance metrics assessed.

Conclusion

The performance of different variable selection methods was compared using 5 performance metrics. Generally, using a higher sample size is recommended when doing variable selection. Under low correlation, the backwards selection methods perform just as good as the shrinkage methods although bias is higher under high correlation for the shrinkage methods. Almost all the methods perform poorly when the effect size is small.

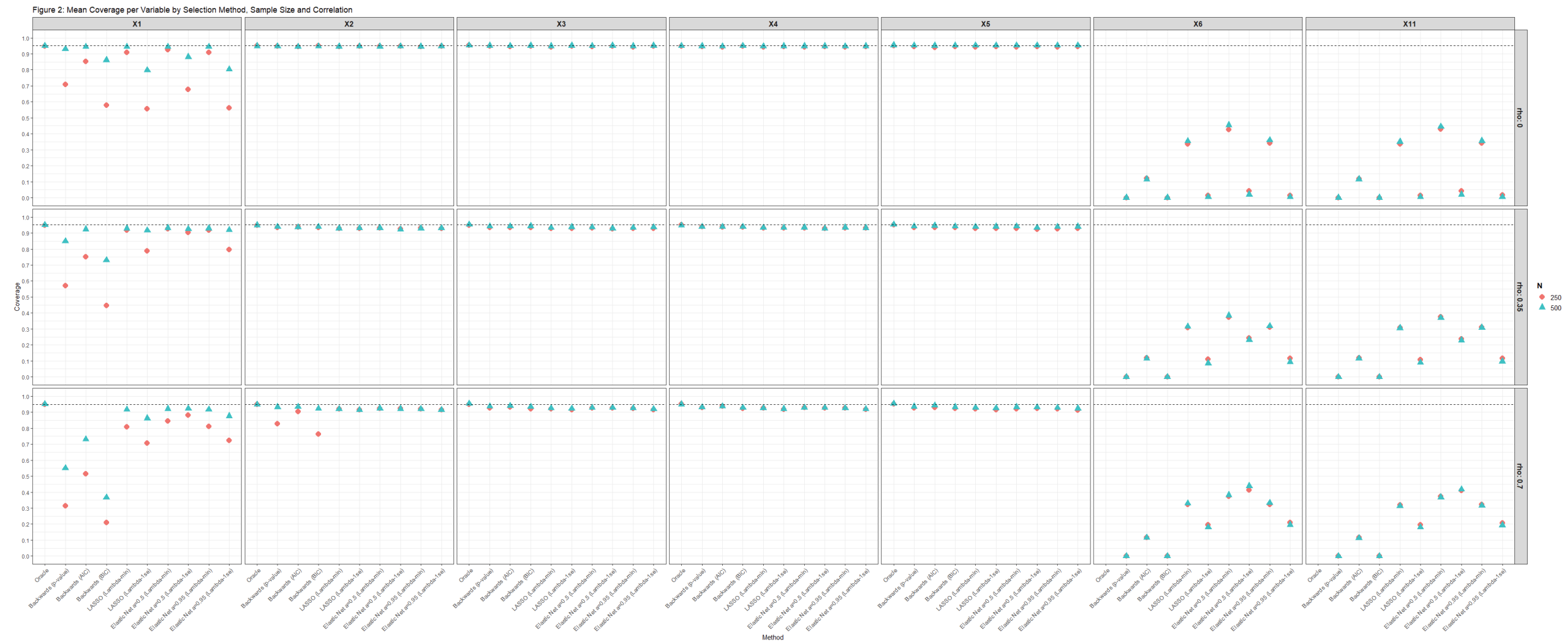
Figure 1: Mean Bias per Variable by Selection Method, Sample Size and Correlation



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Figure 2: Mean 95% Coverage per Variable by Selection Method, Sample Size and Correlation



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Table 1: Mean Bias by Variable Selection Method, Sample Size and Correlation

Variable Selection Method	Rho (ρ)	Sample Size = 250							Sample Size = 500						
		X1	X2	X3	X4	X5	X6	X11	X1	X2	X3	X4	X5	X6	X11
Oracle	0.00	0.000	0.000	0.000	0.000	-0.001	NA	NA	0.000	-0.001	0.000	0.000	0.001	NA	NA
Backwards (p-value)		-0.023	0.000	0.000	0.000	-0.001	0.001	0.000	-0.003	-0.001	0.000	0.000	0.001	0.000	0.000
Backwards (AIC)		-0.007	0.000	0.000	0.000	-0.001	0.001	0.000	-0.001	-0.001	0.000	0.001	0.001	0.000	0.000
Backwards (BIC)		-0.041	0.000	0.000	0.000	-0.001	0.000	0.000	-0.010	-0.001	0.000	0.000	0.001	0.000	0.000
LASSO (Lambda-min)		-0.004	-0.002	-0.001	-0.002	-0.002	0.000	0.000	-0.001	-0.001	0.000	0.000	0.000	0.000	0.000
LASSO (Lambda-1se)		-0.048	-0.003	-0.001	-0.002	-0.002	0.000	0.000	-0.019	-0.001	0.000	0.000	0.000	0.000	0.000
Elastic Net $\alpha=0.5$ (Lambda-min)		-0.003	-0.002	-0.001	-0.002	-0.002	0.000	0.000	-0.001	-0.001	0.000	0.000	0.000	0.000	0.000
Elastic Net $\alpha=0.5$ (Lambda-1se)		-0.031	-0.003	-0.002	-0.003	-0.003	0.000	0.000	-0.009	-0.002	-0.001	0.000	0.000	0.000	0.000
Elastic Net $\alpha=0.95$ (Lambda-min)		-0.004	-0.002	-0.001	-0.002	-0.002	0.000	0.000	-0.001	-0.001	0.000	0.000	0.000	0.000	0.000
Elastic Net $\alpha=0.95$ (Lambda-1se)		-0.047	-0.003	-0.001	-0.002	-0.003	0.000	0.000	-0.019	-0.001	0.000	0.000	0.000	0.000	0.000
Oracle	0.35	0.000	0.000	0.000	0.000	-0.001	NA	NA	0.000	-0.001	0.000	0.001	0.001	NA	NA
Backwards (p-value)		-0.037	0.003	0.004	0.004	0.003	0.002	0.002	-0.010	0.000	0.001	0.002	0.002	0.000	0.000
Backwards (AIC)		-0.014	0.001	0.001	0.001	0.000	0.002	0.001	-0.003	-0.001	0.000	0.001	0.001	0.001	0.000
Backwards (BIC)		-0.057	0.005	0.008	0.008	0.007	0.001	0.001	-0.024	0.002	0.003	0.004	0.004	0.000	0.000
LASSO (Lambda-min)		-0.023	-0.024	-0.024	-0.024	-0.025	0.012	0.011	-0.017	-0.018	-0.016	-0.016	-0.015	0.008	0.008
LASSO (Lambda-1se)		-0.034	-0.025	-0.025	-0.025	-0.026	0.012	0.012	-0.019	-0.017	-0.016	-0.015	-0.015	0.007	0.008
Elastic Net $\alpha=0.5$ (Lambda-min)		-0.021	-0.022	-0.022	-0.023	-0.023	0.011	0.011	-0.015	-0.016	-0.015	-0.015	-0.014	0.007	0.007
Elastic Net $\alpha=0.5$ (Lambda-1se)		-0.034	-0.035	-0.034	-0.035	-0.036	0.016	0.016	-0.025	-0.025	-0.024	-0.024	-0.023	0.011	0.011
Elastic Net $\alpha=0.95$ (Lambda-min)		-0.023	-0.024	-0.024	-0.024	-0.025	0.012	0.011	-0.017	-0.018	-0.016	-0.016	-0.015	0.008	0.008
Elastic Net $\alpha=0.95$ (Lambda-1se)		-0.034	-0.026	-0.026	-0.026	-0.027	0.013	0.012	-0.020	-0.018	-0.017	-0.016	-0.016	0.008	0.008
Oracle	0.70	0.000	0.000	0.001	0.000	-0.001	NA	NA	0.000	-0.002	0.000	0.001	0.002	NA	NA
Backwards (p-value)		-0.066	-0.013	0.011	0.011	0.010	0.004	0.004	-0.039	0.002	0.005	0.006	0.007	0.002	0.001
Backwards (AIC)		-0.030	-0.004	0.003	0.003	0.002	0.003	0.002	-0.016	-0.001	0.001	0.002	0.003	0.001	0.001
Backwards (BIC)		-0.089	-0.021	0.019	0.021	0.020	0.004	0.004	-0.071	0.006	0.013	0.013	0.014	0.001	0.001
LASSO (Lambda-min)		-0.039	-0.048	-0.048	-0.048	-0.049	0.018	0.017	-0.032	-0.035	-0.033	-0.032	-0.031	0.012	0.012
LASSO (Lambda-1se)		-0.051	-0.058	-0.057	-0.058	-0.059	0.022	0.021	-0.040	-0.041	-0.039	-0.038	-0.038	0.014	0.014
Elastic Net $\alpha=0.5$ (Lambda-min)		-0.037	-0.045	-0.045	-0.046	-0.047	0.018	0.017	-0.029	-0.032	-0.031	-0.030	-0.029	0.012	0.011
Elastic Net $\alpha=0.5$ (Lambda-1se)		-0.050	-0.059	-0.059	-0.059	-0.060	0.022	0.021	-0.040	-0.043	-0.041	-0.040	-0.039	0.015	0.015
Elastic Net $\alpha=0.95$ (Lambda-min)		-0.039	-0.048	-0.047	-0.048	-0.049	0.018	0.017	-0.032	-0.034	-0.033	-0.032	-0.031	0.012	0.012
Elastic Net $\alpha=0.95$ (Lambda-1se)		-0.051	-0.059	-0.059	-0.059	-0.060	0.022	0.021	-0.040	-0.042	-0.040	-0.039	-0.039	0.015	0.015

Table 2: Mean Coverage by Variable Selection Method, Sample Size and Correlation

Variable Selection Method	Rho (ρ)	Sample Size = 250							Sample Size = 500						
		X1	X2	X3	X4	X5	X6	X11	X1	X2	X3	X4	X5	X6	X11
Oracle	0.00	0.95	0.95	0.95	0.95	0.95	NA	NA	0.95	0.95	0.95	0.95	0.95	NA	NA
Backwards (p-value)		0.71	0.95	0.95	0.95	0.94	0.00	0.00	0.93	0.95	0.95	0.95	0.95	0.00	0.00
Backwards (AIC)		0.85	0.95	0.94	0.94	0.94	0.12	0.12	0.94	0.94	0.95	0.95	0.95	0.12	0.12
Backwards (BIC)		0.58	0.95	0.95	0.95	0.95	0.00	0.00	0.86	0.95	0.95	0.95	0.95	0.00	0.00
LASSO (Lambda-min)		0.91	0.95	0.94	0.94	0.94	0.34	0.33	0.94	0.95	0.95	0.95	0.95	0.35	0.35
LASSO (Lambda-1se)		0.55	0.95	0.95	0.95	0.95	0.01	0.02	0.80	0.95	0.95	0.95	0.95	0.00	0.00
Elastic Net $\alpha=0.5$ (Lambda-min)		0.92	0.95	0.94	0.94	0.94	0.43	0.43	0.94	0.95	0.95	0.95	0.95	0.45	0.44
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.68	0.95	0.95	0.94	0.94	0.04	0.04	0.88	0.95	0.95	0.95	0.95	0.02	0.02
Elastic Net $\alpha=0.95$ (Lambda-min)		0.91	0.95	0.94	0.94	0.94	0.34	0.34	0.94	0.95	0.95	0.95	0.95	0.36	0.36
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.56	0.95	0.95	0.95	0.95	0.01	0.02	0.80	0.95	0.95	0.95	0.95	0.01	0.00
Oracle	0.35	0.95	0.95	0.95	0.95	0.95	NA	NA	0.95	0.95	0.95	0.95	0.95	NA	NA
Backwards (p-value)		0.57	0.94	0.93	0.94	0.94	0.00	0.00	0.85	0.94	0.94	0.94	0.94	0.00	0.00
Backwards (AIC)		0.75	0.94	0.94	0.94	0.93	0.12	0.12	0.92	0.94	0.94	0.94	0.95	0.12	0.12
Backwards (BIC)		0.45	0.93	0.93	0.94	0.94	0.00	0.00	0.73	0.94	0.94	0.94	0.94	0.00	0.00
LASSO (Lambda-min)		0.92	0.93	0.93	0.93	0.93	0.31	0.31	0.93	0.93	0.93	0.93	0.94	0.32	0.30
LASSO (Lambda-1se)		0.79	0.93	0.93	0.93	0.93	0.11	0.11	0.92	0.93	0.94	0.93	0.94	0.09	0.09
Elastic Net $\alpha=0.5$ (Lambda-min)		0.93	0.93	0.93	0.93	0.93	0.37	0.37	0.93	0.93	0.94	0.94	0.94	0.38	0.37
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.90	0.92	0.93	0.93	0.92	0.24	0.24	0.92	0.92	0.93	0.93	0.93	0.23	0.23
Elastic Net $\alpha=0.95$ (Lambda-min)		0.92	0.93	0.93	0.93	0.93	0.31	0.31	0.93	0.93	0.93	0.93	0.94	0.32	0.31
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.80	0.93	0.93	0.93	0.93	0.12	0.11	0.92	0.93	0.94	0.93	0.94	0.09	0.10
Oracle	0.70	0.95	0.95	0.95	0.95	0.95	NA	NA	0.95	0.95	0.95	0.95	0.95	NA	NA
Backwards (p-value)		0.31	0.83	0.93	0.93	0.93	0.00	0.00	0.55	0.93	0.94	0.93	0.93	0.00	0.00
Backwards (AIC)		0.51	0.90	0.93	0.94	0.93	0.12	0.12	0.73	0.93	0.94	0.94	0.94	0.12	0.11
Backwards (BIC)		0.21	0.76	0.92	0.92	0.92	0.00	0.00	0.37	0.92	0.93	0.93	0.93	0.00	0.00
LASSO (Lambda-min)		0.81	0.92	0.92	0.93	0.92	0.32	0.32	0.92	0.92	0.92	0.93	0.93	0.33	0.31
LASSO (Lambda-1se)		0.71	0.91	0.92	0.92	0.91	0.19	0.19	0.86	0.91	0.92	0.92	0.92	0.18	0.18
Elastic Net $\alpha=0.5$ (Lambda-min)		0.85	0.92	0.93	0.93	0.92	0.37	0.37	0.92	0.92	0.93	0.93	0.93	0.38	0.37
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.88	0.92	0.93	0.93	0.92	0.41	0.41	0.92	0.92	0.93	0.93	0.93	0.44	0.42
Elastic Net $\alpha=0.95$ (Lambda-min)		0.81	0.92	0.92	0.93	0.92	0.32	0.32	0.92	0.92	0.93	0.93	0.93	0.33	0.32
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.72	0.91	0.92	0.92	0.91	0.21	0.21	0.88	0.91	0.92	0.92	0.92	0.20	0.19

Table 3: Rejection Rate by Variable Selection Method, Sample Size and Correlation

Variable Selection Method	Rho (ρ)	Sample Size = 250							Sample Size = 500						
		X1	X2	X3	X4	X5	X6	X11	X1	X2	X3	X4	X5	X6	X11
Oracle	0	0.74	1.00	1.00	1.00	1.00	NA	NA	0.96	1.00	1.00	1.00	1.00	NA	NA
Backwards (p-value)		0.74	1.00	1.00	1.00	1.00	0.05	0.05	0.96	1.00	1.00	1.00	1.00	0.05	0.05
Backwards (AIC)		0.74	1.00	1.00	1.00	1.00	0.06	0.06	0.96	1.00	1.00	1.00	1.00	0.05	0.05
Backwards (BIC)		0.61	1.00	1.00	1.00	1.00	0.03	0.02	0.89	1.00	1.00	1.00	1.00	0.01	0.01
LASSO (Lambda-min)		0.72	1.00	1.00	1.00	1.00	0.05	0.06	0.95	1.00	1.00	1.00	1.00	0.05	0.05
LASSO (Lambda-1se)		0.56	0.99	1.00	1.00	1.00	0.03	0.03	0.82	1.00	1.00	1.00	1.00	0.02	0.02
Elastic Net $\alpha=0.5$ (Lambda-min)		0.72	1.00	1.00	1.00	1.00	0.05	0.06	0.95	1.00	1.00	1.00	1.00	0.05	0.05
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.65	1.00	1.00	1.00	1.00	0.04	0.04	0.90	1.00	1.00	1.00	1.00	0.03	0.03
Elastic Net $\alpha=0.95$ (Lambda-min)		0.72	1.00	1.00	1.00	1.00	0.05	0.06	0.95	1.00	1.00	1.00	1.00	0.05	0.05
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.57	0.99	1.00	1.00	1.00	0.03	0.03	0.83	1.00	1.00	1.00	1.00	0.02	0.02
Oracle	0.35	0.62	0.99	1.00	1.00	1.00	NA	NA	0.89	1.00	1.00	1.00	1.00	NA	NA
Backwards (p-value)		0.60	0.99	1.00	1.00	1.00	0.06	0.06	0.88	1.00	1.00	1.00	1.00	0.05	0.05
Backwards (AIC)		0.60	0.99	1.00	1.00	1.00	0.06	0.06	0.88	1.00	1.00	1.00	1.00	0.06	0.06
Backwards (BIC)		0.47	0.98	1.00	1.00	1.00	0.03	0.02	0.76	1.00	1.00	1.00	1.00	0.01	0.02
LASSO (Lambda-min)		0.47	0.98	1.00	1.00	1.00	0.04	0.04	0.79	1.00	1.00	1.00	1.00	0.04	0.04
LASSO (Lambda-1se)		0.47	0.98	1.00	1.00	1.00	0.01	0.01	0.81	1.00	1.00	1.00	1.00	0.01	0.01
Elastic Net $\alpha=0.5$ (Lambda-min)		0.47	0.98	1.00	1.00	1.00	0.05	0.05	0.79	1.00	1.00	1.00	1.00	0.05	0.05
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.42	0.97	1.00	1.00	1.00	0.01	0.01	0.75	1.00	1.00	1.00	1.00	0.01	0.01
Elastic Net $\alpha=0.95$ (Lambda-min)		0.47	0.98	1.00	1.00	1.00	0.04	0.04	0.79	1.00	1.00	1.00	1.00	0.04	0.04
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.47	0.98	1.00	1.00	1.00	0.01	0.01	0.80	1.00	1.00	1.00	1.00	0.01	0.01
Oracle	0.70	0.36	0.88	1.00	1.00	1.00	NA	NA	0.61	0.99	1.00	1.00	1.00	NA	NA
Backwards (p-value)		0.35	0.87	1.00	1.00	1.00	0.06	0.06	0.58	0.99	1.00	1.00	1.00	0.05	0.06
Backwards (AIC)		0.35	0.86	0.99	1.00	1.00	0.07	0.07	0.58	0.99	1.00	1.00	1.00	0.06	0.06
Backwards (BIC)		0.25	0.81	0.99	1.00	1.00	0.03	0.03	0.40	0.97	1.00	1.00	1.00	0.02	0.02
LASSO (Lambda-min)		0.20	0.72	0.98	1.00	1.00	0.03	0.03	0.40	0.96	1.00	1.00	1.00	0.03	0.03
LASSO (Lambda-1se)		0.17	0.70	0.98	1.00	1.00	0.01	0.01	0.38	0.96	1.00	1.00	1.00	0.01	0.01
Elastic Net $\alpha=0.5$ (Lambda-min)		0.20	0.72	0.98	1.00	1.00	0.04	0.04	0.41	0.96	1.00	1.00	1.00	0.04	0.04
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.15	0.67	0.97	1.00	1.00	0.01	0.01	0.35	0.96	1.00	1.00	1.00	0.01	0.01
Elastic Net $\alpha=0.95$ (Lambda-min)		0.20	0.72	0.98	1.00	1.00	0.03	0.03	0.40	0.96	1.00	1.00	1.00	0.03	0.03
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.17	0.70	0.98	1.00	1.00	0.01	0.01	0.37	0.96	1.00	1.00	1.00	0.01	0.01

Table 4: Mean 95% Coverage by Variable Selection Method, Sample Size and Correlation

Variable Selection Method	Rho (ρ)	Sample Size = 250							Sample Size = 500						
		X1	X2	X3	X4	X5	Prop. X1 – X5	Prop. X6 – X20	X1	X2	X3	X4	X5	Prop. X1 – X5	Prop. X6 – X20
Oracle	0.00	1.00	1.00	1.00	1.00	1.00	1.00	NA	1.00	1.00	1.00	1.00	1.00	1.00	NA
Backwards (p-value)		0.74	1.00	1.00	1.00	1.00	0.95	0.05	0.96	1.00	1.00	1.00	1.00	0.99	0.05
Backwards (AIC)		0.88	1.00	1.00	1.00	1.00	0.98	0.17	0.99	1.00	1.00	1.00	1.00	1.00	0.17
Backwards (BIC)		0.61	1.00	1.00	1.00	1.00	0.92	0.02	0.89	1.00	1.00	1.00	1.00	0.98	0.01
LASSO (Lambda-min)		0.94	1.00	1.00	1.00	1.00	0.99	0.39	1.00	1.00	1.00	1.00	1.00	1.00	0.40
LASSO (Lambda-1se)		0.58	0.99	1.00	1.00	1.00	0.91	0.04	0.82	1.00	1.00	1.00	1.00	0.96	0.02
Elastic Net $\alpha=0.5$ (Lambda-min)		0.96	1.00	1.00	1.00	1.00	0.99	0.48	1.00	1.00	1.00	1.00	1.00	1.00	0.50
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.70	1.00	1.00	1.00	1.00	0.94	0.08	0.91	1.00	1.00	1.00	1.00	0.98	0.05
Elastic Net $\alpha=0.95$ (Lambda-min)		0.94	1.00	1.00	1.00	1.00	0.99	0.40	1.00	1.00	1.00	1.00	1.00	1.00	0.41
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.59	0.99	1.00	1.00	1.00	0.92	0.04	0.83	1.00	1.00	1.00	1.00	0.97	0.03
Oracle	0.35	1.00	1.00	1.00	1.00	1.00	1.00	NA	1.00	1.00	1.00	1.00	1.00	1.00	NA
Backwards (p-value)		0.60	0.99	1.00	1.00	1.00	0.92	0.06	0.88	1.00	1.00	1.00	1.00	0.98	0.05
Backwards (AIC)		0.78	1.00	1.00	1.00	1.00	0.96	0.18	0.95	1.00	1.00	1.00	1.00	0.99	0.17
Backwards (BIC)		0.47	0.98	1.00	1.00	1.00	0.89	0.02	0.76	1.00	1.00	1.00	1.00	0.95	0.02
LASSO (Lambda-min)		0.94	1.00	1.00	1.00	1.00	0.99	0.35	0.99	1.00	1.00	1.00	1.00	1.00	0.35
LASSO (Lambda-1se)		0.80	1.00	1.00	1.00	1.00	0.96	0.12	0.95	1.00	1.00	1.00	1.00	0.99	0.10
Elastic Net $\alpha=0.5$ (Lambda-min)		0.95	1.00	1.00	1.00	1.00	0.99	0.42	1.00	1.00	1.00	1.00	1.00	1.00	0.43
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.92	1.00	1.00	1.00	1.00	0.98	0.25	0.99	1.00	1.00	1.00	1.00	1.00	0.24
Elastic Net $\alpha=0.95$ (Lambda-min)		0.94	1.00	1.00	1.00	1.00	0.99	0.36	0.99	1.00	1.00	1.00	1.00	1.00	0.36
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.81	1.00	1.00	1.00	1.00	0.96	0.13	0.95	1.00	1.00	1.00	1.00	0.99	0.11
Oracle	0.70	1.00	1.00	1.00	1.00	1.00	1.00	NA	1.00	1.00	1.00	1.00	1.00	1.00	NA
Backwards (p-value)		0.35	0.87	1.00	1.00	1.00	0.84	0.06	0.58	0.99	1.00	1.00	1.00	0.91	0.06
Backwards (AIC)		0.55	0.94	1.00	1.00	1.00	0.90	0.18	0.76	1.00	1.00	1.00	1.00	0.95	0.17
Backwards (BIC)		0.25	0.81	0.99	1.00	1.00	0.81	0.03	0.40	0.97	1.00	1.00	1.00	0.87	0.02
LASSO (Lambda-min)		0.82	0.99	1.00	1.00	1.00	0.96	0.35	0.94	1.00	1.00	1.00	1.00	0.99	0.35
LASSO (Lambda-1se)		0.71	0.98	1.00	1.00	1.00	0.94	0.20	0.87	1.00	1.00	1.00	1.00	0.97	0.19
Elastic Net $\alpha=0.5$ (Lambda-min)		0.86	1.00	1.00	1.00	1.00	0.97	0.41	0.95	1.00	1.00	1.00	1.00	0.99	0.41
Elastic Net $\alpha=0.5$ (Lambda-1se)		0.89	1.00	1.00	1.00	1.00	0.98	0.41	0.97	1.00	1.00	1.00	1.00	0.99	0.43
Elastic Net $\alpha=0.95$ (Lambda-min)		0.82	0.99	1.00	1.00	1.00	0.96	0.35	0.94	1.00	1.00	1.00	1.00	0.99	0.36
Elastic Net $\alpha=0.95$ (Lambda-1se)		0.73	0.98	1.00	1.00	1.00	0.94	0.21	0.88	1.00	1.00	1.00	1.00	0.98	0.20

Reproducible Research Information

- GitHub repository for Project 4: <https://github.com/yemiaj/BIOS6624/tree/main/Project4>
- Raw data file (from Canvas) location: *Data was simulated for this project; no raw data was provided.*
- Processed and clean final analysis file location:
“C:\Users\Owner\Documents\cuclasses\spring26\bios_6624\BIOS6624\Project4\DataProcessed\bios6624_p4_sim_results.csv”
 - The .csv file referenced above contains the main simulation results. This was summarized to create the plots and figures in the report.
- Code files layout:
 - **The program files must be run in this sequence to avoid errors.** Each program contains a function called within other functions in the programs
 - *p4_1_data_generation.R*: This code includes a single function to generate a data according to indicated specifics
 - *p4_2_performance_metrics.R*: Includes a single function that computes performance metrics
 - *p4_3_analyze_data.R*: Includes a single function to analyze generated data and perform all variable selection methods
 - *p4_4_expand_parallelize.R*: Parallelizes simulation and runs 10K Monte Carlo samples
 - *p4_5_summarize_results.R*: Includes code to summarize simulation output, tabulate estimates, and creates figures
- Main report
 - “C:\Users\Owner\Documents\cuclasses\spring26\bios_6624\BIOS6624\Project4\Reports\Jegede_Project4_VariableSelectionSimulation.pdf” is the final study report.
- README files in the main and sub-directories were updated with comments to make them more informative.